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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Powered Skill Matching Recruitment System	
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Abstract:

Recruitment teams often need to review many resumes against job descriptions, a process that can be time-consuming and difficult to perform consistently. The AI-Powered Skill Matching Recruitment System is designed to assist recruiters by comparing a candidate's resume or profile with the skills and requirements stated in a job description.

The system accepts candidate resumes and job descriptions as inputs. It identifies information relevant to recruitment, including technical and professional skills, education, experience, projects and certifications, and converts the content into a form suitable for comparison.

Natural language processing is used to clean the text and extract skills and key requirements. Semantic matching helps the system recognize related terms even when the exact wording differs, such as 'Natural Language Processing' and 'NLP'.

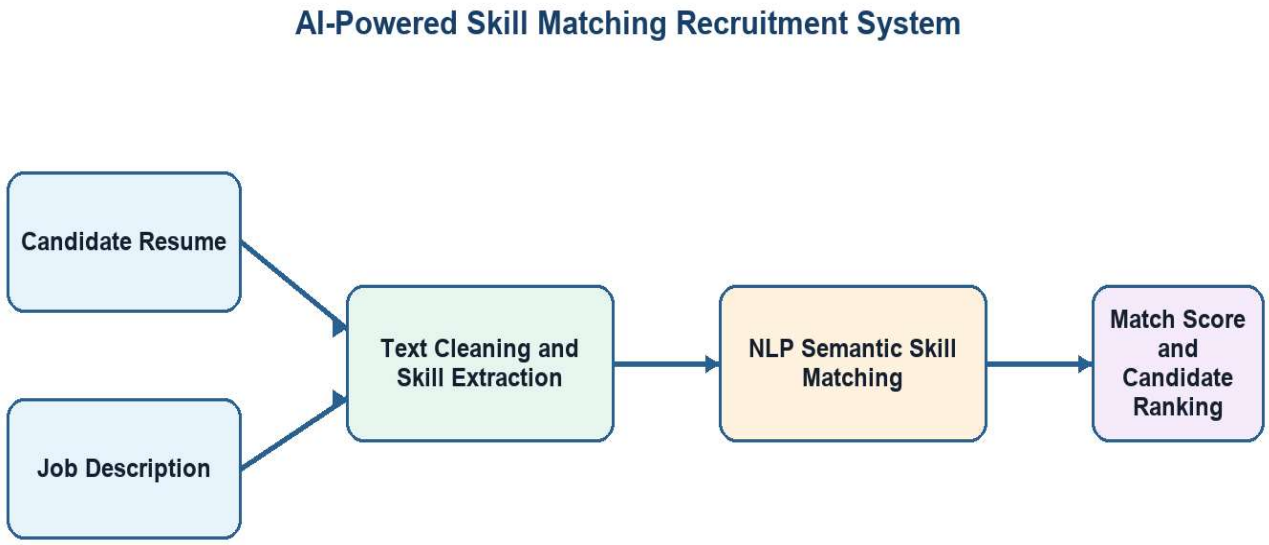
The matching process compares the candidate's available skills with the job's required skills and highlights matching skills, missing skills, relevant experience and overall skill similarity. This makes the result easier for both candidates and recruiters to understand.

An AI-based match score is generated from factors such as skill alignment, required versus available skills, relevant experience, education, certifications and semantic similarity between the resume and the job description. Candidates can then be ranked according to their suitability for the role.

The recruiter dashboard presents the ranked candidates and their match information, enabling recruiters to focus first on profiles that are most relevant to the position. The system supports decision-making and reduces the effort required for initial resume screening.

Overall, the proposed system provides a structured and intelligent approach to recruitment skill matching. By combining resume analysis, job-requirement extraction, semantic comparison and candidate ranking, it aims to improve the efficiency and consistency of the shortlisting process.

Project Architecture/Project Overview:



Recruiter dashboard: matched skills, missing skills, relevant experience and ranked candidates